



PRIORITIZATION OF EMERGING INFECTIOUS DISEASES – BELGIUM, 2025-2026

SUMMARY

- A prioritization exercise was conducted in Belgium focusing on emerging or re-emerging infectious diseases which might constitute a threat for public health in Belgium. The objective of the exercise was to identify priority diseases for which regular risk assessments might be needed in the future to support competent authorities in developing or updating emergency preparedness plans.
- The ECDC-recommended Multi-Criteria Decision Analysis (MCDA) methodology was used.
- Thirty-one diseases, considered as potential (re-)emerging health threats, were assessed against 15 pre-defined criteria covering three domains: A) likelihood of (re-)introduction in Belgium, B) epidemic potential and C) disease impact. The assessment was done through consultation of Belgian experts from different backgrounds and settings.
- Diseases were ranked for each criteria. Ranks per criteria were averaged within each domain (likelihood of (re-)introduction, epidemic potential and disease impact) and then across domains.
- Overall, the top 10 ranked diseases were avian influenza infections, coronaviruses (MERS/SARS/SARS-CoV2), other zoonotic influenza infections, haemorrhagic fevers (Marburg, Crimean-Congo, Ebola and Lassa), West-Nile fever, invasive *Klebsiella* infection and tuberculosis.
- By domain,
 - Avian and zoonotic influenza infections, coronavirus infections and tuberculosis ranked in the top 10 for both likelihood of (re-)introduction and disease impact;
 - West-Nile fever and Chikungunya infection ranked in the top 10 for both likelihood of (re-)introduction and epidemic potential;
 - Haemorrhagic fevers ranked in the top 10 for both epidemic potential and disease impact.

1.1. BACKGROUND

The COVID-19 pandemic has highlighted the vulnerability of our systems to major health threats and the importance of preparedness plans.

In Belgium, a Generic Preparedness Plan (GPP) was developed to strengthen preparedness and response to health crises. The GPP distinguishes between imminent (short term), emerging (medium term), and long-term health risks, and one of the actions was the identification of emerging health risks and threats to public health in Belgium. Health threats can arise from a variety of sources including infectious diseases (due to viral, bacterial, mycological or parasitic agents); infectious diseases due to antimicrobial resistant infectious agents, chemical-biological-radiological or nuclear incidents (CBRN) or environmental events (flood, heat, ..).

The GPP also emphasizes the importance of conducting systematic health risk assessments to enable health authorities to anticipate threats and develop more effective public health policies based on scientific evidence.

The present work is embedded into the GPP framework and was requested by the Belgian health authorities. This work also aligns with the recommendations issued by the ECDC following the Public Health Emergency Preparedness Assessment (PHEPA) conducted in Belgium in 2024 (1). The goal of this work is to provide a ranking of health threats, the scope being placed on (re)-emerging threats and specifically limited to infectious diseases, rather than encompassing all potential public health threats.

The objective of this work is therefore to establish a list of priority emerging or re-emerging infectious diseases which might constitute a threat for public health in Belgium, and for which regular risk assessments might be needed in the future. These risk assessments would support competent authorities in developing or updating emergency preparedness plans (adjust surveillance systems, plan stocks, clarify governance or responsibilities).

WHO defines emerging infectious diseases as diseases that appear in a population for the first time, or that have existed previously but are rapidly increasing in incidence or geographic range. Environmental changes, human and animal demography, pathogen changes and changes in farming practice are among the factors that lead to emerging diseases (2). The American CDC defines emerging infectious diseases as diseases whose incidence in humans has either increased within the past two decades or threatens to increase in the near future (3).

At international level, initiatives developing preparedness plans for emerging infectious diseases are ongoing. WHO is developing PRET (Preparedness and Resilience for Emerging Threats), which aims at providing guidance on pandemic planning (4).

This document first describes the methodology used for the prioritization exercise, then the results are presented and the obtained disease ranking is compared to other priority lists.

1.2. METHODOLOGY – MULTICRITERIA DECISION ANALYSIS

In 2015, the ECDC performed a literature review to identify methods used to prioritize infectious diseases threats for the purpose of emergency preparedness planning, and define best practices for ranking them. Five methodologies were analyzed: bibliometric index, Delphi technique, multi-criteria decision analysis (MCDA), qualitative algorithms and questionnaires (5). Building on this analysis, the ECDC published, in 2017, a tool for the prioritization of infectious diseases threats (6). The ECDC tool was based on the MCDA, as it was shown to assist ranking in a transparent, comparable and methodologically reproducible manner. The method was originally derived from a methodological risk ranking approach described by Havelaar et al. for emerging zoonoses (7).

In line with the recommendation of ECDC, the current prioritization of diseases was carried out using the MCDA. The key elements of the MCDA-based risk prioritization process are as follows:

1. Planning: determine scope and purpose of the ranking exercise
2. Identify diseases for prioritization
3. Formulate a list of criteria against which to assess diseases
4. Weight criteria
5. Score diseases according to criteria
6. Rank diseases based on relative scores

1.2.1. Planning: determine scope and purpose of the ranking exercise

A team comprising four people from Sciensano (services Epidemiology of Infectious Diseases, Crisis Preparedness and Response and Risk Assessment Group) was established to coordinate the prioritization exercise. This Coordination team predefined the objectives and the scope of the exercise, and identified the experts to be consulted in the subsequent steps.

A first expert group was set up, bringing together specialists from Sciensano representing several services and covering a broad range of expertise ("Expert Group 1" described in Annex 1). The objective of the exercise, the scope and the definitions were discussed and finalized with this group.

A search of the literature was conducted on previous prioritization exercises undertaken in Europe and internationally, to inform the current exercise (summarized in Annex 2).

The ECDC Emergency Preparedness and Response team was consulted in spring 2025 for feedback on the methodology.

1.2.2. Identify diseases for prioritization

Based on previous prioritization exercises, the Coordination team aimed for a list of around 30 diseases.

- 1- A first list of diseases was produced combining the initial lists of diseases used for the exercises carried out by Klamer et al. (8), Ploin et al (9) and the European Food Safety Agency (EFSA) (10) -> **140 diseases**
- 2- The list was first reduced by the Coordination team, removing diseases not representing a potential (re)-emerging public health threat for Belgium -> **70 diseases**

- 3- The Expert Group 1 was consulted during the summer of 2025 through a Limesurvey questionnaire: experts were asked to specify whether, in their opinion, each of the 70 diseases constituted a potential (re)-emerging threat for public health in Belgium or not. The “I don’t know” option was also given. Eleven out of 24 experts replied to the survey. Diseases were ranked according to the number of “Yes” responses. The result of the survey and the ranking are available in Annex 3. The Coordination team reviewed the list and agreed on a selection of **31 diseases**, based on the top-ranked diseases, literature review and their own expert judgement.
- 4- MERS, SARS, COVID-19 and Coronaviruses other than COVID-19, MERS and SARS were grouped into “Coronaviruses” because the characteristics and impact of these diseases in Belgium was considered comparable.
- 5- Six diseases were added after discussion with Sciensano experts and consultation of ECDC documents: Marburg virus disease, Rift valley Fever, Brucellosis, Leptospirosis, Hantavirus disease and Nipah virus disease.
- 6- Three diseases were removed after discussion with additional Sciensano experts: Mumps and Lyme disease were finally not retained as emerging threats to public health in Belgium and Anthrax was considered out of the scope of this exercise as being more of a CBRN threat.

The final list of diseases identified for the prioritization exercise is shown in Table 1.

Table 1: List of diseases identified for the prioritization exercise (in alphabetical order)

	diseases
1	Brucellosis
2	Chikungunya disease
3	Coronaviruses with human spillover potential (MERS, SARS, COVID-19)
4	Crimean-Congo hemorrhagic fever
5	Dengue
6	Diphtheria
7	Ebola virus disease
8	Hantavirus infection
9	Influenza avian
10	Influenza human
11	Influenza zoonotic other than avian
12	Invasive Staphylococcus aureus
13	Invasive <i>Klebsiella</i>
14	Lassa fever
15	Leptospirosis
16	Malaria
17	Marburg virus disease
18	Measles
19	Mpox
20	Neisseria gonorrhoeae disease
21	Nipah virus disease
22	Pertussis
23	Polio
24	Q fever
25	Rift valley fever
26	Tick-borne encephalitis
27	Tuberculosis
28	Vibrio cholerae and parahaemolyticus diseases
29	West Nile virus disease
30	Yellow fever
31	Zika virus disease

1.2.3. Formulate a list of criteria against which to assess diseases

A list of possible criteria based on Klamer et al (8) and Ploin and al (9) was presented to the Expert Group 1 during a meeting held in the spring of 2025. After discussion with the expert group, 15 criteria were retained and grouped into three main domains: A) Likelihood of introduction/(re-)emergence; B) Epidemic potential and C) Impact of the disease. The criteria were then further refined with the same Expert Group 1 during a second in-person workshop held in June 2025, and through email consultation over the summer of 2025. The criteria are shown in Table 2 and detailed in Annex 4.

- The **Likelihood of introduction/(re)emergence** reflects the probability that the disease is introduced or reintroduced into the human population in Belgium in the next five years, it includes the amount of drivers facilitating (re)emergence/introduction as well as the likelihood of animal to human transmission.
- The **Epidemic potential** refers to the ability of a disease to spread in the Belgian population, it includes the human to human transmission pattern, the immunity in the Belgian population, the adaptability of the disease agent as well as the different public health measures available (preventive, curative, non-pharmaceutical and diagnostics).
- The **Impact of the disease** describes the effects the disease has on individuals, communities, and the health care system in Belgium. The socio-economic impact of the diseases is not considered in the scope of this exercise.

Table 2 : Identified criteria for the prioritisation exercise

A- Likelihood of introduction/(re-)emergence	A1- Likelihood of introduction in the human population
	A2- Drivers of (re-) emergence
	A3- Expected incidence in the human population in the next 5 years
	A4- Likelihood animal-to-human transmission
B- Epidemic potential	B1- Likelihood human-to-human transmission
	B2- Population immunity
	B3- Intrinsic adaptability of the disease agent
	B4- Preventive countermeasures available
	B5- Curative countermeasures available
	B6- Availability of non-pharmaceutical interventions
	B7- Capability of early detection
C- Impact of the disease	C1- Impact on human health – case fatality
	C2- Impact on human health - morbidity
	C3- Impact in health care system
	C4- Impact on vulnerable populations

1.2.4. Weight criteria according to importance

In this exercise, all criteria were considered of equal importance and no weighting was applied. This approach is supported by previous work by ECDC and Africa CDC, which showed that weighting criteria within domains had very limited impact on the final rank for almost all diseases considered (11).

1.2.5. Score diseases against criteria

1.2.5.1. Experts assessment via questionnaire

A questionnaire was prepared on Limesurvey to assess the 31 diseases against the 15 criteria (questions and answer options detailed in Annex 4). A first version was presented to the Expert Group 1 at the workshop in June 2025, where experts piloted the questionnaire and provided feedback on the structure and clarity of the questions. The questionnaire was designed so that respondents answered each question across all 31 diseases before moving on to the next question.

The final version of the questionnaire was sent to 132 experts in Belgium, members of the Risk Assessment Group or the National Reference Centers. This group covered a broad range of expertise across different fields, settings and regions of Belgium. The affiliation, domain of expertise and professional role of respondents were recorded. The option was given to answer anonymously.

The questionnaire provided background information on the diseases to be ranked, including the current incidence in Belgium for each disease in the human population and, when relevant, in the animal population.

The survey was distributed to the 132 experts by email on October 1st 2025 and remained open until Mid-November 2025.

1.2.5.2. Scoring

The answers of each expert were given a score as shown in Table 3. For questions A4, B1 and B3, the answer “Not Applicable” was given the lowest score as it was predominantly selected for question–disease combinations in which the aspect of introduction risk (A) or epidemic potential (B) addressed by the question did not contribute to the introduction risk or epidemic potential of the disease concerned. For questions A1 and A2, the interpretation of the “Not Applicable” answer was less straightforward, so these answers were treated as missing in the analysis. Answers “I don’t know” (DK) were also treated as missing for all questions.

Table 3: Scoring the answers provided by the experts (NA= Not Applicable; DK= I don't know)

Domain	Q	Answers and scores						Treated as missing	
Likelihood of introduction	A1	Very unlikely	Unlikely	Likely	Very likely			NA	DK
		1	2	3	4				
	A2	Climate change	AMR	Transport	Trade	Tourism	Other	NA	DK
		1	1	1	1	1	1		
	A3	Yes	No						DK
		4	1						
	A4	Not applicable	Low	Medium	High				DK
		1	2	3	4				
Epidemic potential	B1	Not applicable	Low	Medium	High				DK
		1	2	3	4				
	B2	None	Low	Medium	High				DK
		4	3	2	1				
	B3	Not applicable	Slow	Fast					DK
		1	2	3					
	B4	None	Low	Medium	High				DK
		4	3	2	1				
	B5	None	Low	Medium	High				DK
		4	3	2	1				
	B6	None	Low	Medium	High				DK
		4	3	2	1				
	B7	None	Low	Medium	High				DK
		4	3	2	1				
Impact of the disease	C1	Low	Medium	High	Very high				DK
		1	2	3	4				
	C2	Low	Medium	High	Very high				DK
		1	2	3	4				
	C3	Low	Medium	High	Very high				DK
		1	2	3	4				
	C4	Low	Medium	High	Very high				DK
		1	2	3	4				

1.2.6. Rank diseases based on relative scores

1.2.6.1. Main analysis

Since experts were not required to answer all questions, the number of missing responses ("I don't know" or "Non Applicable" depending the question) was not expected to be equally distributed across experts, questions and diseases. To account for the variable number of missing answers across experts and questions, a mixed model approach was developed per question, with disease as a fixed effect and expert as a random intercept.

Three model types were used depending on the response structure of the question:

- (a) For the 12 questions with ordinal response scales of 4 levels (A1, A4, B1, B2, B4–B7, C1–C4), a cumulative link mixed model (CLMM) was fitted using the ordinal R package. For B3 (3 ordinal levels: not applicable / slow / fast), a CLMM with 3 levels was fitted.
- (b) For A3 (binary: yes/no), a binomial generalized linear mixed model (GLMM) was fitted.
- (c) For A2 (count of drivers selected), a Poisson GLMM was fitted.

For each model, disease-specific coefficients (log-odds or log-rate ratios relative to a reference disease) were estimated, and diseases were ranked per question by these coefficients, with higher values indicating higher priority.

Per-question ranks were subsequently averaged within each question domain (Likelihood of introduction, Epidemic potential and Impact of the disease). The overall ranking was then obtained by averaging the average ranks of each question domain.

One disease-question combination (Ebola, question B4) was excluded from the analysis due to a data recording error in Limesurvey.

1.2.6.2. Sensitivity analysis

The main analysis was carried out on all available responses (including both partially and fully answered questionnaires). For the sensitivity analysis, the model described above was applied only to the answers of respondents who completed the whole questionnaire.

The modelling was performed by statisticians from Sciensano, service Epidemiology of Infectious Diseases (see Annex 1).

1.3. RESULTS

1.3.1. Objective of the exercise and definitions

As delineated through discussions with the Sciensano Expert Group 1, the **purpose** of this exercise was to **prioritize infectious diseases that are considered as emerging or re-emerging threats to health for Belgium, and for which regular risk assessment will be needed in the future**. Those regular risk assessments will support competent authorities in updating emergency preparedness plans.

The notion of emerging infectious diseases in the context of this exercise was discussed with the Expert Group 1 and was defined as follows:

Emerging infectious diseases are diseases that either (i) appear in a population for the first time, (ii) that have existed previously but are increasing in incidence or geographic range (both gradual and rapid increase) or (iii) that threaten to increase in the coming five years and cause a public health crisis in Belgium.

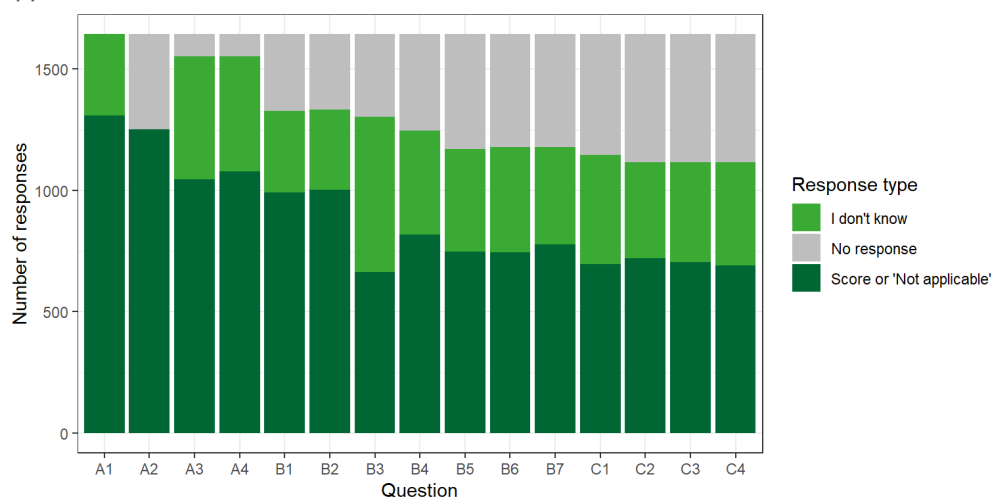
The focus of this prioritization exercise is on infectious diseases that affect humans. Infectious diseases affecting animals with a minimal impact on human health are out of the scope of this exercise. Biological agents considered as CBRN are also out of the scope of this exercise.

1.3.2. Descriptive analysis

1.3.2.1. Responses

A total of 53 experts responded to at least one question (response rate 40,1%). After filtering out empty records and duplicate responses, a list of 1643 individual responses (53 experts x 31 diseases) was obtained for each of the 15 questions, including both valid and missing responses. Two patterns were observed: (1) the number of valid responses decreased towards the end of the questionnaire, with 36 out of 53 experts completing all questions, whereas 17 discontinued participation before completion; and (2) there was a substantial number of “I don’t know” answers, varying per question (Figure 1).

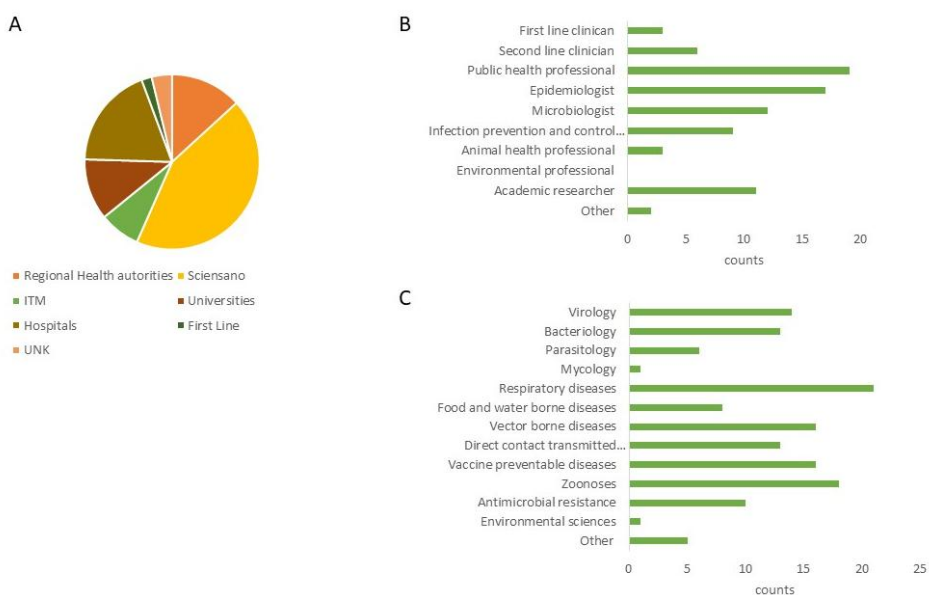
Figure 1: Counts of response types across all survey questions and diseases. Green: question answered “I don’t know”; Grey: question not answered; Dark green: question given a scoreable answer or “Not applicable”



1.3.2.2. Respondents profile

Figure 2 shows the profile of the 53 experts who participated to the questionnaire responding to at least one question. The most represented affiliation among respondents was Sciensano (43 %), followed by Belgian hospitals (19 %), regional health authorities (13 %), universities (11 %), the Institute of Tropical Medicine (8 %), and primary care physicians (2 %), covering a wide range of expertise and professional roles.

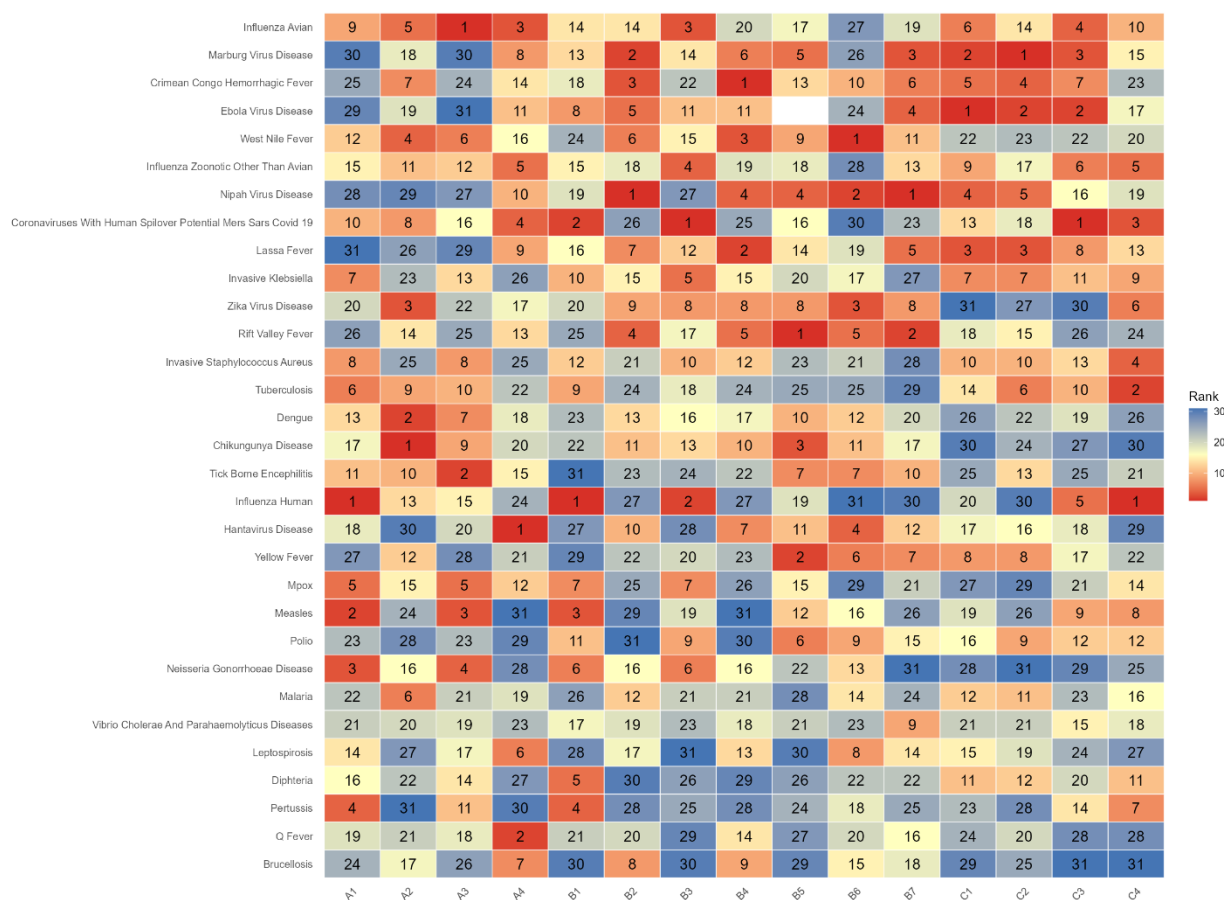
Figure 2 Respondent profile. A- Affiliation ; B- Professional role; C- Domain of expertise (ITM = Institute of Tropical Medicine, UNK = unknown).



1.3.3. Ranking

Figure 3 shows the ranking matrix of each disease and each question obtained from the cumulative-linked mixed model.

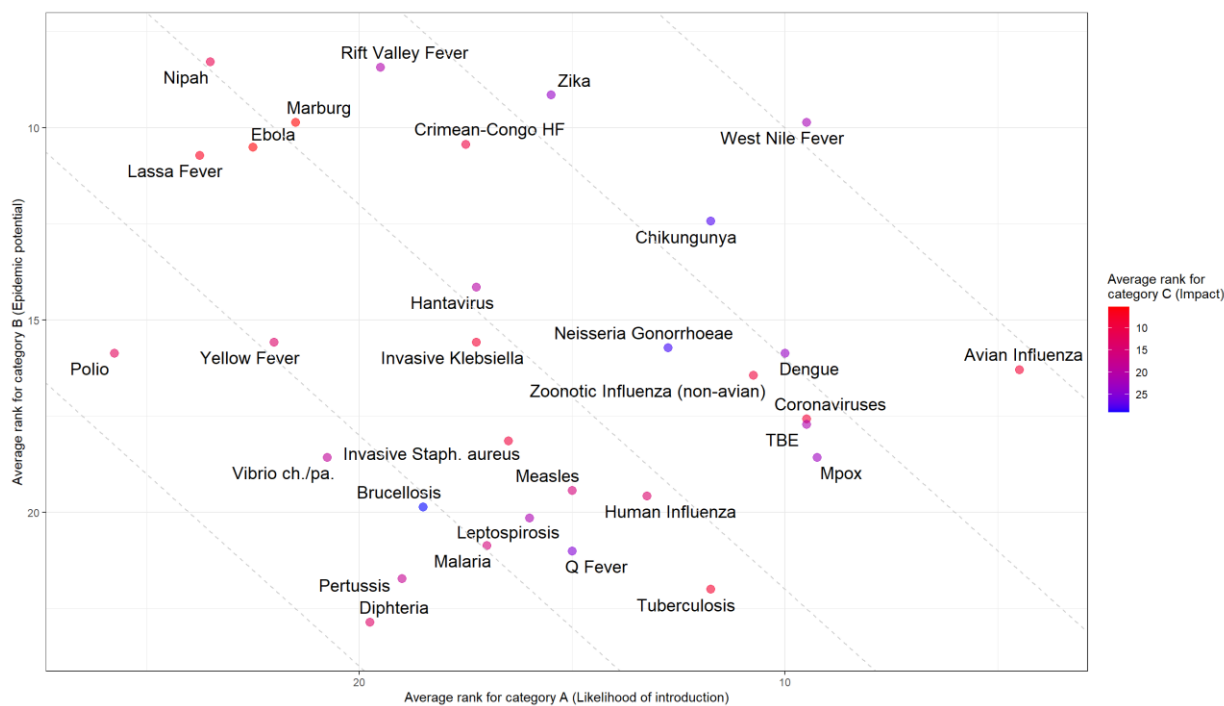
Figure 3: Disease ranking per question – Cumulative-Linked Mixed Model (CLMM). Heatmap showing the rank of each disease (y axis) for each question (x axis - questions A1 to C4). For each question, a colour gradient is applied to the ranking positions (first ranking positions (red) to last ranking positions (blue)).



1.3.3.1. Disease ranking per question domain

For each disease, an average rank was calculated for each question domain (Likelihood of introduction, Epidemic potential and Impact of the disease). Figure 4 displays the average ranks by domain in three dimensions. Diseases with the highest priority ranked in the first positions and appear on the right of the x-axis (Likelihood of introduction), on the top of the y-axis (Epidemic potential) and in red colour (Impact of the disease). The average ranks by domain are also presented in Annex 5.

Figure 4: Disease ranking by question domain. Scatter plot combining the ranking of each disease on Likelihood of introduction (x-axis), Epidemic potential (y-axis) and Impact of the disease (colour gradient). The isolines show points with equal value for the average rank for Likelihood of introduction and Epidemic potential.



Considering the domain **Likelihood of introduction**, the ten diseases with the highest average ranks included zoonotic influenza (avian and other), coronaviruses, several vector-borne diseases (tick-borne encephalitis, West-Nile fever, dengue, chikungunya), as well as mpox, tuberculosis and gonorrhoea. In contrast, polio, Nipah virus disease and some haemorrhagic fevers (Lassa fever, Ebola and Marburg) showed the lowest average ranks.

For the domain **Epidemic potential**, West-Nile fever and chikungunya appeared again among the top-ranked diseases, alongside Zika, haemorrhagic fevers (Marburg, Ebola, Crimean-Congo fever, Lassa fever), as well as Nipah virus disease, hantavirus disease and Rift valley fever. The lowest average ranks were observed for diphtheria, tuberculosis, pertussis, Q fever and malaria. Finally, considering the domain **Impact of the disease**, the haemorrhagic fevers ranked high again (Marburg, Ebola, Crimean-Congo fever, Lassa fever) as well as zoonotic influenza (avian and other), coronaviruses, tuberculosis and invasive bacterial infections (*Klebsiella* and *Staphylococcus*). On the other hand, brucellosis, gonorrhoea, chikungunya, Zika and Q fever presented the lowest average ranks.

1.3.3.2. Overall disease ranking

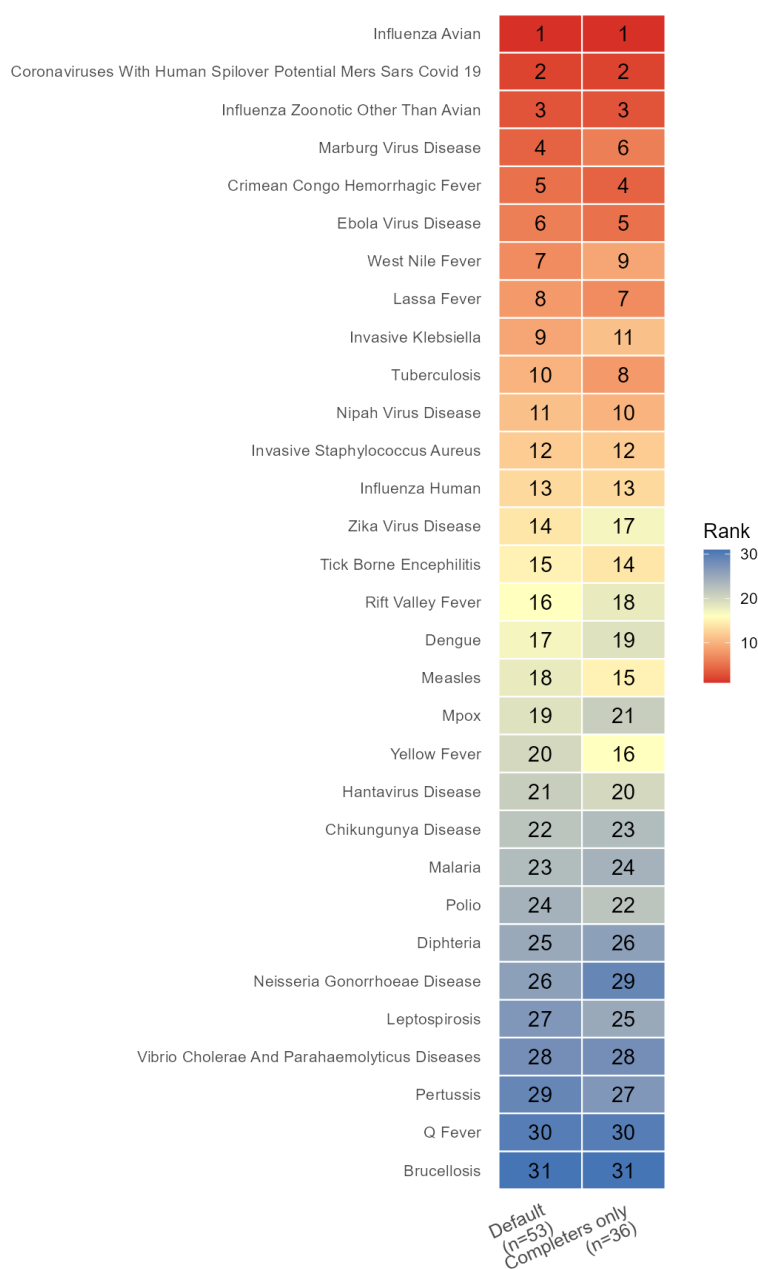
The overall average rank, calculated by averaging the average rank of each disease in each question domain, is shown in Figure 5 (left ranking). The diseases with the highest rank were zoonotic influenza (avian and other), coronaviruses, haemorrhagic fevers (Marburg, Ebola, Crimean-Congo and Lassa fevers), West-Nile fever, invasive *Klebsiella* infection and tuberculosis.

The diseases with the lowest overall average ranks were brucellosis, Q fever, pertussis, Vibrio infections and leptospirosis.

1.3.3.3. Sensitivity analysis

The same methodology was applied considering only the responses from experts who replied to the entire questionnaire (n=36). The overall ranking results, shown in Figure 5 (right ranking), are comparable to the main analysis.

Figure 5: Overall disease ranking. **Default:** Cumulative Linked Mixed Model ranks averaged within each question domain (Likelihood of introduction, Epidemic potential and Impact of the disease), then equally weighted across question domains, responses from all experts who responded to at least one question are included (n=53); **Completers only:** same method restricted to the responses of the 36 experts who replied to all questions.



1.4. COMPARISON OF THE OBTAINED RESULTS WITH OTHER PRIORITY LISTS

- The prioritization exercise carried out in France in 2024, with the objective of establishing a list of priority infectious diseases for public health, surveillance and research in mainland and overseas France, also identified viral haemorrhagic fevers (rank 1), mosquito-borne arboviruses (rank 3), zoonotic influenza (rank 4), emerging coronaviruses (rank 6) and tuberculosis (rank 13) as high priorities (9). In their high priority group, and not identified in our exercise, were “Disease X”, several respiratory infections (seasonal influenza A and B, RSV and hMPV and other acute respiratory infections), Creutzfeldt–Jakob disease, systemic infections due to MDR bacteria, infections due to emerging XDR bacteria, invasive infections due to *Neisseria meningitidis*, rabies and invasive pneumococcal disease. On the other hand, infections due to hypervirulent clonal strains of *Klebsiella pneumoniae* was in the low-priority group (rank 46) while it ranked in the top 10 in the present study. The criteria used in the French prioritization exercise were comparable to the criteria used in the present study, but they were weighted, with incidence and case fatality rates obtaining 25 % of the total weighting points.
- The results obtained here are also aligned with WHO, which prioritized diseases for research and development in emergency contexts, based on international epidemiological criteria. Priority diseases included viral haemorrhagic fevers, respiratory infections and mosquito-borne arbovirus diseases (12).
- The exercise carried out by EFSA in 2022 focused on zoonoses, the aim was to identify priorities for the establishment of a coordinated surveillance system under the One Health approach. A list was established at EU level (see Annex 2) but also at country level (10). Of note, for Belgium, zoonotic influenza (avian and swine) had also ranked in the top priorities, as well as COVID-19 and Crimean-Congo haemorrhagic fever.
- The prioritization exercise carried out in Belgium in 2018 identified different priorities than the present study (pertussis, HIV-infection, hepatitis B and C, Rubella, influenza, Tuberculosis, Invasive candidiasis and Mumps), but the objectives were very different. In 2018, the aim of the prioritization exercise was to rank communicable diseases in order to inform allocation of resources for public health action in Belgium, while the current exercise focuses on (re-)emerging infectious diseases to inform preparedness.

1.5. LIMITATIONS OF THE STUDY

- The participation rate was acceptable (40,1 %), but 17/53 participating experts (32%) dropped out before completion of the survey. This may be related to the long questionnaire, which required time to assess 31 diseases against 15 criteria. Other methodologies might be considered to increase expert engagement, such as the Delphi method or pairwise comparisons in which experts are asked to compare two diseases at a time.
- The availability of the “I don’t know” answer option allowed experts to abstain from assessing diseases outside of their area of expertise, enabling participation of a wider range of experts. However, this increased the risk of bias because some disease-question combinations were assessed by a smaller number of experts. This was partially addressed with the modelling approach that was chosen, but residual biases might

remain. The number of experts for some disease-question combinations was limited, however each still received a minimum of seven answers.

- The interpretation of emerging and re-emerging diseases might have been different among experts. A definition was provided, which had been discussed beforehand with the Expert Group 1. However some uncertainties remained regarding diseases that are already endemic in Belgium but could, under certain circumstances, be considered as a potential threat. Emerging diseases and re-emerging diseases might have been considered separately.
- The initial reduction of the disease list from 140 to 70 diseases was carried out by the Coordination team alone, based on expert judgement and prior to the formal consultation process. As such, diseases excluded at that stage were not evaluated using the same systematic criteria applied in subsequent steps. This approach was chosen for practical reasons, to reduce the burden on experts, by first removing diseases that clearly did not represent a potential (re-)emerging public health threat for Belgium.
- For this exercise, the choice was made to assign equal weights to each question within a question domain, then equal weights to each question domain on the overall ranking. This methodological choice was made based on previous work by the ECDC and Africa CDC; it also reduced the burden on experts, by avoiding an additional consultation step to assign weights to criteria. It would, however, be valuable to explore whether experts might assign greater importance to certain criteria than others, for the purpose of this exercise.
- In this exercise, with the exception of Coronaviruses, diseases were assessed individually rather than grouped into broader categories (e.g., viral hemorrhagic fevers, vector-borne diseases). This approach allowed for more granular assessments, and was chosen because diseases in a same group were considered as having different transmission routes, seasonal patterns, public health implications and thus different preparedness and response strategies. However, not grouping diseases might have influenced the composition of the final ranking, as diseases belonging to the same group occupy multiple positions in the top ranks, potentially at the expense of diseases from other groups. Assessing individual diseases also contrasts with WHO recent prioritization exercise for which a pathogen-family approach was chosen (13). This approach considers that a novel infectious disease threat could arise from a new pathogen emerging from an already existing pathogen family. Although the present study provides a ranking for diseases caused by pathogens that are considered as prototypes for several pathogens families, we acknowledge that preparedness efforts should also reflect a wider family approach.
- The interpretation of the “Not Applicable” answer option was not always straightforward, which led to a different scoring depending on the question. For questions A4, B1 and B3, the answer “Not Applicable” was given the lowest score, while for questions A1 and A2, it was treated as missing in the analysis. Ideally, “Not Applicable” answers should be handled consistently across all questions in future exercises.
- Uncertainty in the log-odds estimation was not propagated to the ranking, and therefore diseases with similar average ranks should not be considered meaningfully different.

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1.7. ANNEXES

Annex 1 Sciensano experts constituting Expert Group 1 and Coordination team

Name	Service
Giulietta Stefani	Epidemiology Infectious Diseases - RAG
Javiera Rebolledo	
Laura Cornelissen	
Tinne Lernout	
Koen Blot	
Dieter Van Cauteren	Epidemiology Infectious Diseases
Joris Van Loenhout	
Laurane De Mot	
Claire Brugerolles	
Natalia Bustos Sierra	
Veronik Hutse	Health services research
Nathalie Bossuyt	
Boudewijn Catry	Healthcare-associated infections and antimicrobial resistance
Carlotta Di Bari	Health information
Steven Van Gucht	Infectious diseases in humans
Koenraad Vanhoorde	Foodborne pathogens
Sarah Denayer	Viral diseases
Sanne Terryn	
Mieke Steensels	Avian virology and immunology
Marcella Mori	Veterinary bacteriology
Marylene Tignon	Viral reemerging enzootic and bee diseases
Mickael Cargnon	Veterinary epidemiology
Simon Couvreur	Statistical analyses - Epidemiology Infectious Diseases - RAG
Toon Braeye	Statistical analyses - Epidemiology Infectious Diseases
Géraldine De Muylder	Coordination team - Epidemiology Infectious Diseases – RAG
Géraldine Boseret	Coordination team – Crisis Preparedness & Response
Clara Mazagatos Ateca	Coordination team – Statistical analyses - EPIET fellow
Jorgen Stassijns	Coordination team - Crisis Preparedness & Response – RAG

Annex 2 Prioritization exercises undertaken in Europe and internationally

Study	Method	Result	Ref
Prioritization of livestock transboundary diseases in Belgium using a multicriteria decision analysis tool based on drivers of emergence (2015)	MCDA	29 livestock diseases exotic to Belgium were prioritized based on drivers. Porcine epidemic diarrhoea, foot-and-mouth disease, low pathogenic avian influenza, African horse sickness and highly pathogenic avian influenza were identified as priorities.	(14)
Prioritization for future surveillance, prevention and control of 98 communicable diseases in Belgium: a 2018 multi-criteria decision analysis study	MCDA	98 diseases were scored for 18 criteria and classified into 5 priority groups. The diseases ranked as very high priorities were pertussis, HIV-infection, hepatitis C, hepatitis B, Rubella, influenza, Tuberculosis, Invasive candidiasis and Mumps.	(8)
Pathogen prioritization for wastewater surveillance ahead of the Paris 2024 Olympic and Paralympic games, France	Delphi	Six priority targets suitable for wastewater surveillance during the Olympics were identified (poliovirus, influenza A and influenza B viruses, mpox virus, SARS-CoV-2 and measles virus).	(15)
Prioritization of infectious diseases from a public health perspective: a multi-criteria decision analysis study, France, 2024	MCDA	15 infectious entities were ranked as high-priority (Disease X ¹ , viral hemorrhagic fevers, ARIs ² , mosquito-borne arboviruses, zoonotic influenza virus infections, seasonal influenza A and B, RSV and hMPV, Creutzfeldt–Jakob disease, MDR bacteria infections, XDR bacteria infections, Neisseria meningitidis infections, rabies, tuberculosis and invasive pneumococcal disease), 32 were low-priority and 48 were classified as non-priority.	(9)
WHO pathogen prioritization: a scientific framework for epidemic and pandemic research preparedness, 2024	Delphi (Pathogen Family approach)	Twelve pathogen families were considered as presenting a high risk, two families as a medium risk and 16 a low or low to medium risk.	(13)
EFSA: prioritisation of zoonotic diseases for coordinated surveillance systems under the One Health approach for cross-border pathogens that threaten the Union, 2022	Combination of MCDA and Delphi	A list of ten priorities and Disease X ¹ were considered relevant for the preparation of specific surveillance strategies: Crimean-Congo hemorrhagic fever, echinococcosis, hepatitis E, influenza (avian), influenza (swine), Lyme borreliosis, Q-fever, Rift Valley fever, tick-borne encephalitis, and West Nile fever.	(10)
Africa: risk ranking and prioritization of epidemic-prone diseases, 2022	Combination of MCDA and Delphi	Diseases with the highest ranks were Ebola, Cholera and COVID-19 whilst monkeypox, anthrax, plague and Chikungunya had the lowest ranks.	(11)
UK: priority pathogen families research and development (R&D) tool - a reference tool to help guide England-based funders of research and development	Delphi (Pathogen Family approach)	The priority pathogen family R&D tool contains a list of 16 viral families and 8 bacterial families. These families are not ranked.	(16)

MCDA: Multi-Criteria Decision Analysis; HIV: Human Immunodeficiency Virus, SARS-CoV-2: Severe Acute Respiratory Syndrome Coronavirus 2, RSV: Respiratory Syncytial Virus, hMPV: human Metapneumovirus, MDR: Multi-Drug Resistant, XDR: Extensively drug-resistant, COVID-19: Coronavirus Disease 2019

¹ Disease X is a possible future, but still unknown, emerging disease

² Acute Respiratory Infections due to viruses other than influenza, emerging coronaviruses, Respiratory Syncytial Virus (RSV) and human Metapneumovirus (hMPV).

Annex 3 Selection of diseases for the prioritization exercise, Summer 2025.

Result of a Limesurvey consultation of the Expert Group 1. Experts were asked to specify whether, in their opinion, each of the 70 diseases constituted a potential (re)-emerging threat for public health in Belgium or not.

Diseases	YES	NO	I don't know	no answer
[Dengue]	11	0	0	0
[Chikungunya disease]	10	0	1	0
[Influenza (avian)]	10	1	0	0
[West Nile virus disease]	10	0	1	0
[Measles]	9	2	0	0
[Middle East Respiratory Syndrome (MERS)]	9	1	1	0
[Mpox]	9	1	1	0
[Tick-borne encephalitis]	9	0	2	0
[Crimean-Congo hemorrhagic fever]	8	2	1	0
[Dyphtheria]	8	1	2	0
[Influenza]	8	3	0	0
[Influenza (swine)]	8	3	0	0
[Severe Acute Respiratory Syndrome (SARS)]	8	3	0	0
[Coronavirus diseases (other than COVID-19, MERS or SARS)]	7	2	2	0
[COVID-19]	7	2	2	0
[Pertussis]	7	2	2	0
[Polio]	7	3	1	0
[Zika virus disease]	7	2	2	0
[Invasive Staphylococcus aureus]	6	2	3	0
[Lyme disease]	6	5	0	0
[Mumps]	6	3	2	0
[Tuberculosis]	6	2	3	0
[Anthrax]	5	3	3	0
[Ebola virus disease]	5	6	0	0
[Invasive <i>Klebsiella</i>]	5	1	5	0
[Lassa fever]	5	6	0	0
[Malaria]	5	5	1	0
[Neisseria gonorrhoeae disease]	5	4	2	0
[Q fever]	5	3	3	0
[Vibrio cholerae and parahaemolyticus diseases]	5	4	2	0
[Yellow fever]	5	4	2	0
[Invasive Aspergillosis]	4	2	5	0
[Invasive Candidiasis]	4	1	6	0
[Leishmaniasis]	4	4	3	0
[Syphilis]	4	5	2	0
[Tularemia]	4	4	3	0
[Invasive Enterococci]	3	2	6	0

[Lyssavirus infections (other than rabies)]	3	3	5	0
[Marburg virus disease]	3	7	0	1
[Nipah virus disease]	3	4	4	0
[Plague]	3	7	1	0
[Poxvirus disease]	3	4	4	0
[Rift valley fever]	3	3	4	1
[Rubella]	3	5	3	0
[Shigellosis]	3	5	3	0
[Smallpox]	3	5	3	0
[Usutu virus disease]	3	1	7	0
[Anaplasmosis]	2	5	3	1
[Arenavirus disease (other than Lassa fever)]	2	3	6	0
[Brucellosis]	2	3	5	1
[Hantavirus infection]	2	5	2	2
[Japanese encephalitis]	2	5	3	1
[Orthobunyavirus disease]	2	2	7	0
[Rickettsia diseases (other than spotted fevers)]	2	3	6	0
[St Louis encephalitis]	2	3	6	0
[Babesiosis]	1	6	4	0
[Eastern equine encephalitis]	1	4	6	0
[Glanders (Burkholderia mallei)]	1	5	5	0
[Helvetica spotted fever]	1	3	7	0
[Invasive Acinetobacter]	1	3	7	0
[Pseudomonas disease]	1	7	3	0
[Rabies]	1	9	1	0
[Sindbis fever]	1	3	7	0
[Western equine encephalitis]	1	3	7	0
[Hendra virus disease]	0	3	8	0
[Mediterranean spotted fever]	0	2	9	0
[Pappataci fever]	0	2	9	0
[Powassan virus disease]	0	2	9	0
[Shuni virus disease]	0	2	9	0
[Venezuelan equine encephalitis]	0	4	7	0

Annex 4 Detailed description of the criteria and questions included in the questionnaire
The questionnaire was designed so that respondents answered each question across all 31 diseases before moving on to the next question.

Criteria	Possible answers
A1- Likelihood of introduction How would you evaluate the probability that the following diseases could be introduced (i.e. enter for the first time) or reintroduced (i.e. return after being eliminated) into the human population in Belgium in the next 5 years? When making your assessment consider different pathways of introduction such as travel, migration, trade of live animals, animal movements, wildlife trafficking, etc..	Very unlikely Unlikely Likely Very likely Not applicable
A2- Drivers of emergence/re-emergence Which factors would drive the emergence or re-emergence of the following diseases in Belgium? Please add an "x" to all drivers that, in your opinion, apply. Note that you have the possibility to specify which "other" drivers apply by writing in the corresponding box. Emergence is defined as the appearance of a new disease in the population. Re-emergence is defined as the return of a disease that was previously under control or eliminated.	Climate change Antimicrobial resistance Transport Trade Tourism Other Not applicable I don't know
A3- Expected incidence Do you expect an increase in the incidence of the following diseases in the human population in Belgium in the next five years?	Yes No I don't know
A4- Likelihood animal-to-human transmission What is the likelihood of animal-to-human transmission for the following diseases? This question addresses the intrinsic transmissibility, between animals and humans, of the pathogen causing the disease. <i>Not applicable</i> = No known animal-to-human transmission <i>Low</i> = Transmission requires direct contact with infected animals and/or a high infectious dose is needed to cause disease <i>Medium</i> = Transmission can occur through close (but not necessarily direct) contact with infected animals and/or a moderate infectious dose can cause disease <i>High</i> = Transmission may occur over a distance (e.g., airborne, environmental exposure), and/or a low infectious dose is sufficient to cause disease.	Not applicable Low Medium High I don't know
B1- Likelihood human-to-human transmission What is the likelihood of human-to-human transmission for the following diseases, after introduction into the human population in Belgium? <i>Not applicable</i> = No known human-to-human transmission <i>Low</i> = Transmission is rare and inefficient, and requires close or prolonged contact. There can be sporadic cases, not resulting in outbreaks <i>Medium</i> = Transmission can occur under certain conditions or in defined settings, and can occasionally result in small-scale/localised outbreaks. Sustained spread is unlikely <i>High</i> = Transmission is efficient between humans and sustained spread is likely, causing large-scale outbreaks	Not applicable Low Medium High I don't know
B2- Population immunity How would you evaluate the existing immunity against the following diseases in the Belgian human population? Consider both natural and vaccine-induced immunity in the general population at this point in time in Belgium. <i>No immunity</i> = Disease is new or never circulated; no vaccine available or used <i>Low</i> = Immunity exists only in small, specific groups (e.g. travelers, lab workers, older adults). The majority of the population is susceptible <i>Medium</i> = Some groups in the population are immune (e.g. due to partial vaccination programs or past outbreaks), but significant gaps remain (e.g. by age, region, risk group) <i>High</i> = Widespread immunity due to routine vaccination or extensive past exposure; only small or well-characterized groups may lack protection	No immunity Low Medium High I don't know
B3- Disease agent evolution and adaptability How would you rate the evolution capacity and adaptability of the pathogen causing each disease? That is, how quickly it can mutate or evolve in ways that increase its ability to	Not applicable

infect or spread among humans? Consider factors like mutation rate, recombination ability, adaptability to new hosts, and past evidence of rapid evolution (e.g. zoonotic jumps or immune escape).

Not applicable = The pathogen does not evolve or mutate in a way relevant to emergence and spread in humans

Slow = The pathogens evolves gradually, there is little evidence of rapid adaptation or change

Fast = The pathogen shows high mutation or recombination rate, it adapts rapidly to new hosts or environments

Slow

Fast

I don't know

B4- Preventive countermeasures available

Are there any preventive countermeasures available for the human population and ready for use in Belgium for the following diseases? By preventive countermeasures we mean pharmaceutical interventions such as vaccines or any other preventive treatment (Prep, RSV mAb). Non pharmaceutical interventions (masks, isolation, contact tracing) are not covered here, and will be addressed in a question below.

None = No preventive countermeasures, or very limited in number with low effectiveness, are available in Belgium in a timely manner

Low = A limited number of preventive countermeasures are available in Belgium in a timely manner and/or their effectiveness is moderate

Medium = A limited number of highly effective preventive countermeasures are available in Belgium in a timely manner or a sufficient number of preventive countermeasures are available in Belgium in a timely manner but their effectiveness is limited

High = Sufficient and effective preventive countermeasures are available in Belgium in a timely manner

None

Low

Medium

High

I don't know

B5- Curative countermeasures available

Are there curative treatments (excluding symptomatic or supportive care) available and ready to use in Belgium, for the human population, for the following diseases? Consider the effectiveness of the treatment in reducing morbidity and mortality in affected individuals, as well as its acceptability and the potential existence of resistance to the treatment.

None = No treatment exists to reduce morbidity and/or mortality, or a high/moderate/low effective treatment exists but it is not available in a timely manner in Belgium or it has a low acceptability

Low = A treatment is available in Belgium in a timely manner but its effectiveness is low and/or antimicrobial resistance has been documented

Medium = Moderate effective treatment exists and is available in a timely manner, or highly effective treatment exists but availability is limited

High = Highly effective treatment exists and is available in a timely manner

None

Low

Medium

High

I don't know

B6- Availability of non-pharmaceutical interventions

Are there any non-pharmaceutical interventions (NPIs) available and feasible for the human population to help contain the transmission of the following diseases? NPIs include physical distancing, masks, mosquito nets, ... Feasibility includes the effectiveness and the acceptability of such interventions to affected communities.

None = NPIs are not available or they are available but would have low effectiveness or compliance

Low = NPIs are partially available and feasibility is considered to be limited

Medium = NPIs are available and moderately feasible, or they very feasible but partially available

High = NPIs are available and very feasible

None

Low

Medium

High

I don't know

B7- Diagnostic capacity

How would you evaluate the diagnostic capacity to detect the pathogens causing the following diseases in the human population, in Belgium? Consider the existence of relevant diagnostic methods, as well as their timely availability, and their effectiveness to be used during a public health emergency response (including upscaling).

None = No diagnostics exist or no standard diagnostics are available in Belgium

Low = Diagnostics availability is limited in Belgium and/or their effectiveness is limited

Medium = Effective diagnostics exist but their availability in Belgium is limited, or diagnostics are available but their effectiveness is limited

High = Effective diagnostics exist and they are available in Belgium

None

Low

Medium

High

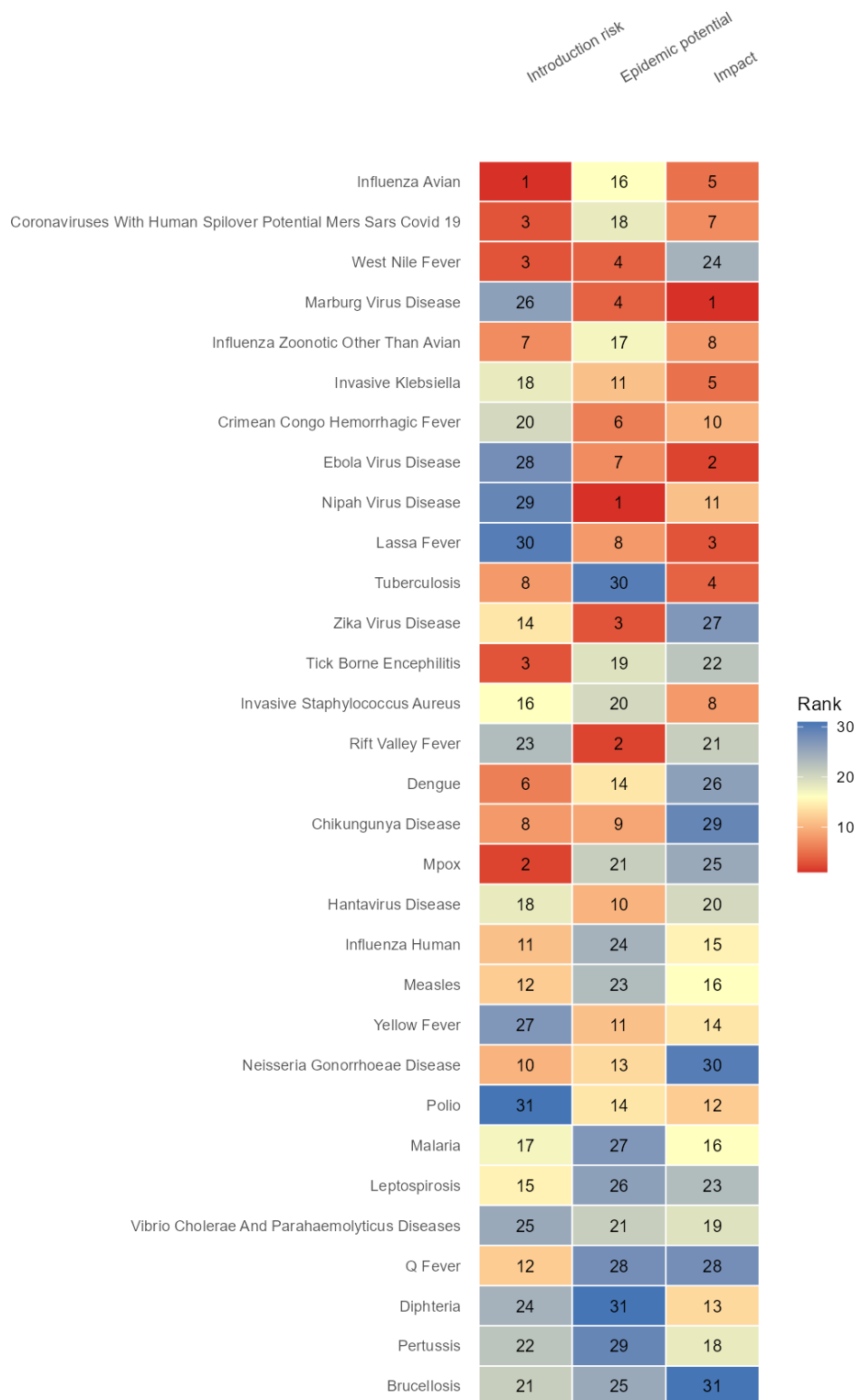
I don't know

C1- Impact on human health - case fatality

Low

How would you estimate the case fatality rate of the following diseases? (proportion of people who die from the disease among all confirmed cases)	Medium
<i>Low</i> = <0,01% if untreated (if treatment available)	High
<i>Medium</i> = 0,01-1% if untreated and/or <0,01% if treated	Very high
<i>High</i> = >1% if untreated and/or 0.01-1% if treated	
<i>Very high</i> = >1% if treated	I don't know
C2- Impact on human health - morbidity	
How would you evaluate the impact of the following diseases on morbidity at the individual level (physical and psychological consequences of the disease other than death) for the majority of the cases in Belgium?	None
<i>None</i> = No or mild symptoms, with no long-term consequences	Low
<i>Low</i> = Mild symptoms with short-term health effects (<1 month) and/or no long-term consequences	Medium
<i>Medium</i> = Symptoms may require medical intervention or hospitalisation, and/or persisting health effects (>1 month) can occur	High
<i>High</i> = Medical intervention or hospitalisation is required and/or persisting health effects (>1 month) are common	I don't know
C3- Impact on healthcare systems	
How would you evaluate the impact of the following diseases on healthcare systems (GP practices, hospitals, healthcare workers) in Belgium? Please assess the impact in a situation where the disease has been introduced in Belgium and has caused a major outbreak. Consider factors such as GP consultations, hospitalisation rates, length of stay in hospital and potential risks of transmission to healthcare workers.	None
<i>None</i> = The disease is responsible for a negligible use of healthcare services	Low
<i>Low</i> = The disease causes a small number of consultations, with limited strain on healthcare services	Medium
<i>Moderate</i> = The disease causes a moderate number of consultations or hospitalisations, potentially affecting resource availability	High
<i>High</i> = The disease causes a substantial pressure on healthcare services, including high demand for care and/or risks to healthcare workers	I don't know
C4- Impact on vulnerable populations	
How would you evaluate the impact of the following diseases on vulnerable populations in Belgium, assuming the disease has been introduced? Vulnerable populations include children, pregnant women, elderly, or other specific populations such as nursing home residents, undocumented individuals, minorities or people with at risk behaviours.	None
<i>None</i> = No specific risk or severity observed in vulnerable groups	Low
<i>Low</i> = Vulnerable groups may experience slightly more severe outcomes, but the overall risk remains low	Medium
<i>Medium</i> = Vulnerable groups are at increased risk of complications, requiring targeted public health attention	High
<i>High</i> = The disease disproportionately affects vulnerable populations with severe outcomes	I don't know

Annex 5 Average ranking of diseases within each question domain (CLMM)



Disease ranking per category (equal weight per question)